AI VIET NAM – AI Conquer 2026

# A Comparative Study of Tree-based Models for Healthcare Risk Prediction: Cross-validation and SHAP-based Stability Analysis

Group **Numby**

| **Name** | **Position** | **Role** |
|---|---|---|
| Le Quang Thanh | Tech Leader | Team Leader |
| Nguyen Van Vi | AI Engineer (Pipeline) | Member |
| Le Thanh Lam | AI Engineer (Data) | Member |
| Pham Minh Dang Tran | AI Engineer (Model) | Member |
| Tan Du | QA / Reviewer | Member |


# I. Research Summary

# Contents

# II. Introduction and Research Questions

## II.1. Research Context

Healthcare risk prediction has become an important application of machine learning, particularly with the increasing availability of structured healthcare data. Machine learning models can be used to identify patterns in demographic, clinical, and physiological features and estimate the risk of developing a disease or experiencing an adverse health outcome. Such predictive models may provide useful decision-support information for healthcare-related analysis.

Among machine learning approaches for structured data, tree-based models are widely used because they can capture nonlinear relationships and feature interactions while requiring relatively few assumptions about the underlying data distribution. In particular, boosting-based models such as AdaBoost, XGBoost, and LightGBM have demonstrated strong predictive capabilities and are commonly applied to classification problems.

However, healthcare datasets can differ substantially in terms of sample size, feature distributions, missing values, and class imbalance. As a result, the performance of a particular model may vary across datasets. Evaluating multiple models across multiple healthcare datasets under a consistent experimental protocol can therefore provide a more comprehensive understanding of their behavior.

## II.2. Research Problem

Although tree-based models can achieve strong predictive performance, comparing models based solely on a single train-test split or a single performance metric may provide an incomplete view of their reliability. A model that performs well on one particular split may exhibit considerable variation when trained on different subsets of the data.

Cross-validation provides a systematic approach for evaluating this variability by repeatedly training and evaluating models on different data partitions. Therefore, model comparison should consider not only average predictive performance but also the consistency of performance across cross-validation folds.

Furthermore, predictive performance does not indicate why a model produces its predictions. In healthcare-related machine learning, identifying influential features can provide additional insight into model behavior. SHapley Additive exPlanations (SHAP) can be used to quantify the contribution of individual features to model predictions. However, feature importance obtained from a single training run may not necessarily remain consistent when the training data changes.

This raises a second stability concern: whether the explanations produced by a model remain consistent across different cross-validation folds. A feature that appears highly important in one fold but substantially decreases in importance in other folds may indicate that the corresponding interpretation is sensitive to the training data.

Therefore, this study considers both predictive stability and explanation stability when comparing tree-based models for healthcare risk prediction.

## II.3. Research Questions

Based on the research problem described above, this study investigates the following research questions:

- **RQ1:** How do AdaBoost, XGBoost, and LightGBM compare in healthcare risk prediction across multiple datasets?
- **RQ2:** How stable is the predictive performance of these models across cross-validation folds?
- **RQ3:** How stable are SHAP-based feature importance rankings across cross-validation folds?

## II.4. Contributions

This study makes the following contributions:

- **Systematic comparison of boosting-based tree models.** We provide a controlled empirical comparison of AdaBoost, XGBoost, and LightGBM for healthcare risk prediction across multiple datasets using a consistent experimental protocol.
- **Evaluation of predictive stability.** In addition to reporting predictive performance, we analyze the variability of model performance across cross-validation folds to assess the consistency of the models.
- **Analysis of SHAP explanation stability.** We investigate the stability of SHAP-based feature importance rankings across cross-validation folds, providing an empirical assessment of whether model explanations remain consistent under different training subsets.
- **Multi-dataset evaluation.** By evaluating the models across multiple healthcare datasets, the study examines whether observed performance and explanation patterns are consistent across different data characteristics.
- **Reproducible experimental framework.** The experiments are conducted using a standardized and reproducible pipeline, enabling consistent comparison across models, datasets, and cross-validation folds.

# III. Related Work

## III.1. Boosting-Based Tree Ensembles

Boosting constructs an ensemble sequentially so that later learners focus on errors made by earlier learners. This makes boosting well suited to structured tabular prediction because non-linear effects and feature interactions can be captured without requiring a single highly complex tree. The three models used in this study represent different generations of boosting-based tree methods: AdaBoost provides a classical boosting baseline, while XGBoost and LightGBM are modern gradient-boosted decision-tree systems designed for stronger predictive performance and computational efficiency.

## III.2. AdaBoost

AdaBoost is one of the foundational boosting algorithms. It updates observation weights after each boosting round, increasing emphasis on samples that were previously misclassified, and combines the weak learners into a weighted final classifier [1]. In this study, AdaBoost serves as the classical baseline and uses decision stumps as weak learners. Keeping the base learners simple provides a clear comparison with the deeper and more heavily regularized gradient-boosting systems used by XGBoost and LightGBM.

## III.3. XGBoost

XGBoost extends gradient tree boosting with both statistical regularization and systems-level optimizations for scalable learning. Its design includes a regularized objective, sparsity-aware split handling, approximate tree learning, and efficient parallel computation [2]. It also exposes controls such as tree depth, shrinkage, row subsampling, column subsampling, and regularization. These controls are useful for the present study because the three healthcare datasets differ substantially in sample size, dimensionality, and class prevalence, while the same fixed model configuration must remain reproducible across datasets.

## III.4. LightGBM

LightGBM is a gradient-boosted decision-tree framework designed for high computational efficiency. Ke et al. introduced Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to reduce computational cost while preserving predictive quality on large and high-dimensional datasets [3]. LightGBM also grows trees leaf-wise, allowing it to fit complex nonlinear relationships efficiently. Because leaf-wise growth can increase model complexity, parameters such as the number of leaves, maximum depth, minimum child samples, learning rate, subsampling, and regularization are controlled explicitly in the experimental configuration.

## III.5. Explainability and Stability of Tree Models

Predictive performance alone does not show which variables drive model outputs. SHAP (SHapley Additive exPlanations) provides additive feature attributions for individual predictions and establishes desirable consistency properties for feature attribution [4]. Tree-specific SHAP algorithms make these explanations computationally practical for tree ensembles, and prior work has demonstrated their use in medical prediction settings [5]. This motivates evaluating not only predictive performance across cross-validation folds but also whether feature-importance rankings remain consistent as the training sample changes.

## III.6. Research Gap and Motivation

The goal of this study is therefore broader than identifying a single model with the highest average score. AdaBoost, XGBoost, and LightGBM are evaluated under a common preprocessing, cross-validation, metric, and hold-out testing protocol across multiple healthcare datasets. The study then examines two forms of stability: fold-to-fold variation in predictive performance and fold-to-fold variation in SHAP-based feature rankings. This design supports the three research questions by separating average predictive strength from sensitivity to the particular training sample and from the stability of model explanations.

# IV. Methodology and Research Architecture

# V. Data and Experimental Design

## V.1. Model Experimental Design

The modeling stage consumes the processed development and test files produced by the upstream data pipeline. For each of the three datasets, the data pipeline creates a stratified 80/20 development–test split. The test partition is held out during model comparison and is used only after cross-validation has been completed. All three candidate algorithms receive the same processed feature matrix for a given dataset so that differences in results are attributable to the model rather than to different input representations.

Within the development partition, we apply stratified five-fold cross-validation with shuffling and a fixed random seed of 42. Stratification preserves approximately the same class proportion in each fold, which is particularly important when the positive healthcare outcome is less frequent [6]. The exact same fold-generation rule and random seed are used for AdaBoost, XGBoost, and LightGBM. For every fold, four folds are used for model fitting and the remaining fold is used for validation. Model objects are re-created from scratch for every fold to prevent fitted state from leaking between folds.

To reduce the effect of class imbalance during fitting, balanced sample weights are computed from the training portion of each fold only. Validation and test sets remain untouched so that the reported metrics reflect the natural class prevalence of each dataset. A fixed probability threshold of 0.50 is used for the threshold-dependent metrics. No threshold is optimized on the validation folds, which avoids adding a second tuning procedure that could make fold comparisons less consistent.

## V.2. Model Configuration and Hyperparameters

The core comparison uses one fixed, literature-guided configuration per algorithm across all three datasets. We intentionally do not select a different best hyperparameter configuration for each dataset using the same five folds that are later used to quantify stability. Doing so could make the reported cross-validation scores optimistic and would confound RQ2 with hyperparameter search variability. The selected settings use conservative learning rates, moderate ensemble sizes, and regularization/subsampling for the gradient- boosted models.

| Model | Selected hyperparameters | Rationale |
|---|---|---|
| AdaBoost | Decision stump (max depth = 1); 200 estimators; learning rate = 0.10 | Decision stumps are the standard weak learner for AdaBoost. The larger ensemble combined with a smaller learning rate provides gradual boosting while limiting the contribution of each individual stump. |
| XGBoost | 300 estimators; learning rate = 0.05; max depth = 4; subsample = 0.80; column sample = 0.80; $L_2$ regularization = 1; histogram tree method | A low learning rate and moderate tree depth control model complexity. Row and column subsampling reduce variance, while $L_2$ regularization makes leaf weights more conservative. The histogram method improves efficiency on the larger datasets. |
| LightGBM | 300 estimators; learning rate = 0.05; 31 leaves; max depth = 6; minimum child samples = 20; subsample = 0.80; column sample = 0.80; $L_2$ regularization = 1 | The number of leaves and depth cap constrain the leaf-wise tree growth. The minimum child size, subsampling, and regularization reduce overfitting while the small learning rate supports a smoother boosting process. Deterministic CPU settings and a fixed seed are enabled for reproducibility. |

Table 1: Fixed model configurations used for the cross-dataset comparison.

## V.3. Evaluation Metrics and Model Comparison

Each validation fold is evaluated using ROC-AUC, PR-AUC, Recall, and F1-score. ROC-AUC measures ranking performance across classification thresholds. PR-AUC focuses on the precision–recall trade-off and is especially informative when the positive class is relatively rare [7]. Recall measures the proportion of positive cases correctly identified, while F1-score balances precision and recall at the fixed 0.50 threshold. Because the three datasets may have different class prevalence, PR-AUC is used as the primary ranking metric, with ROC-AUC, F1-score, and fold variability retained for a more complete comparison.

For RQ1, we report the mean five-fold score of every metric for every model–dataset pair and rank the three models within each dataset by mean PR-AUC. We additionally report the average within-dataset rank across all three datasets rather than pooling patient rows across datasets. This prevents the largest dataset from dominating the overall comparison.

For RQ2, predictive stability is measured directly from the five fold-level scores. For each metric we report the mean, standard deviation, minimum, maximum, and range across folds. A smaller standard deviation and range indicate that the model is less sensitive to the particular development-data partition. After cross-validation, each model is re-fitted once on the full development partition and evaluated on the untouched test partition. These test results are reported separately from the cross-validation results and are not used to select the best model.

The complete experiment configuration is stored in `config.yaml`. The implementation writes fold-level metrics, aggregated cross-validation summaries, held-out test metrics, per-dataset rankings, cross-dataset rankings, and an experiment metadata file containing package versions and a hash of the configuration. This makes the model experiments repeatable and provides the fold-level outputs required by the later SHAP stability analysis. “‘latex

## V.4. SHAP Explainability and Feature-Ranking Stability (RQ3)

To address RQ3, this study uses SHAP (SHapley Additive exPlanations) to examine both the importance of individual input features and the stability of these explanations across cross-validation folds. SHAP assigns an additive contribution value to each feature for an individual prediction and is grounded in Shapley-value principles from cooperative game theory [4]. For tree-based models, TreeSHAP provides an efficient method for estimating these feature attributions [5].

The SHAP analysis follows the same predefined stratified five-fold partitions used in the predictive experiments. These fold assignments are generated by the data-preprocessing pipeline and stored in `data/processed/<dataset>/kfold_indices.csv`. Reusing the same folds ensures that predictive stability and explanation stability are evaluated under identical train–validation partitions.

For each dataset, model, and fold, the model is trained on four folds and SHAP values are computed on a deterministic sample from the held-out validation fold. This procedure prevents the explained observations from being reused as training samples and produces one independent feature-importance ranking for each fold. The random seed, maximum number of explained validation samples, background size, and top-$k$ value are controlled through `config.yaml` to ensure reproducibility.

For XGBoost and LightGBM, SHAP values are computed using `TreeExplainer`. The scikit-learn implementation of AdaBoost used in this study is not supported by TreeExplainer in the pinned SHAP version. Therefore, AdaBoost is explained using model-agnostic permutation SHAP with a deterministic background sample drawn from the corresponding fold-training data. Because the underlying explainer differs between AdaBoost and the gradient-boosted models, raw SHAP magnitudes are not compared directly across model families. Instead, the stability analysis focuses on how each model's own feature ranking changes across folds.

For a feature $j$ in fold $f$, global SHAP importance is defined as the mean absolute SHAP value across the $N_f$ explained validation observations:

$$I_{j,f} = \frac{1}{N_f} \sum_{i=1}^{N_f} |\phi_{i,j,f}|, \tag{1}$$

where $\phi_{i,j,f}$ denotes the SHAP attribution of feature $j$ for observation $i$ in fold $f$. Mean absolute SHAP is used because positive and negative contributions should not cancel when measuring overall feature importance. Features are sorted in descending order of $I_{j,f}$ to construct a fold-specific feature ranking.

To evaluate the robustness of these rankings, three complementary stability metrics are used. First, Kendall's rank correlation coefficient $\tau$ measures the agreement between two feature rankings based on concordant and discordant feature pairs. Values closer to one indicate stronger agreement in pairwise feature ordering.

Second, Spearman's rank correlation coefficient $\rho$ measures monotonic agreement between the complete rank vectors of two folds. A value close to one indicates that the overall ordering of features remains highly consistent between folds.

Third, stability of the most influential features is measured using top-$k$ Jaccard similarity:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}, \tag{2}$$

where $A$ and $B$ represent the sets of the top-$k$ ranked features from two cross-validation folds. The experiment uses $k = 10$. A Jaccard score of one indicates identical top-10 feature membership, whereas a lower score indicates that some features enter or leave the top-10 across folds.

With five cross-validation folds, ten unique fold pairs are evaluated for every dataset–model combination. Kendall's $\tau$ and Spearman's $\rho$ assess stability of the complete feature ranking, while top-10 Jaccard similarity focuses specifically on the features considered most important.

A consensus ranking is additionally constructed by aggregating fold-level results. For each feature, the analysis records its mean absolute SHAP value, average rank, rank standard deviation, and frequency of appearing among the top-10 features. This provides a summary of both average feature importance and the degree to which individual features maintain their importance across different data partitions.

Several visual analyses are generated to complement the numerical stability metrics. SHAP beeswarm plots visualize the distribution and direction of feature effects across held-out validation observations, while mean absolute SHAP bar plots summarize global feature importance. Violin summary plots provide an additional view of the distribution of SHAP contributions for the most important features.

For stability analysis, feature-ranking heatmaps display the position of the leading features across the five folds. These plots make rank changes directly visible and help identify features whose importance is sensitive to the training partition. Fold-correlation heatmaps visualize the pairwise Spearman correlation between fold-specific feature rankings, making it possible to identify individual folds that differ from the remaining partitions.

The implementation stores fold-level SHAP importance values, pairwise stability measurements, aggregated stability statistics, consensus rankings, and the generated explainability and stability plots. The corresponding empirical results are presented in Section VI.3.. "‘

# VI. Results and Discussion

## VI.1. Predictive Performance Across Datasets (RQ1)

Table 2 summarizes the five-fold cross-validation performance of AdaBoost, XGBoost, and LightGBM on the three healthcare datasets. The values are reported as mean ± standard deviation across the five validation folds. PR-AUC is treated as the primary comparison metric, while ROC-AUC, Recall, and F1-score provide complementary views of ranking and threshold-based performance.

On Dataset 1, XGBoost and LightGBM produced almost identical ROC-AUC, while XGBoost achieved a slightly higher mean PR-AUC and Recall. Both gradient- boosted models clearly

| Dataset | Model | ROC-AUC | PR-AUC | Recall | F1 |
|---|---|---|---|---|---|
| Dataset 1 | AdaBoost | $0.8716 \pm 0.0042$ | $0.3853 \pm 0.0087$ | $0.6794 \pm 0.0106$ | $0.3765 \pm 0.0082$ |
| | XGBoost | $0.8872 \pm 0.0033$ | $0.4138 \pm 0.0084$ | $0.7782 \pm 0.0071$ | $0.3324 \pm 0.0026$ |
| | LightGBM | $0.8873 \pm 0.0033$ | $0.4129 \pm 0.0088$ | $0.7771 \pm 0.0084$ | $0.3346 \pm 0.0031$ |
| Dataset 2 | AdaBoost | $0.9241 \pm 0.0419$ | $0.9226 \pm 0.0444$ | $0.8449 \pm 0.0294$ | $0.8578 \pm 0.0190$ |
| | XGBoost | $0.9244 \pm 0.0322$ | $0.9214 \pm 0.0324$ | $0.8815 \pm 0.0381$ | $0.8785 \pm 0.0249$ |
| | LightGBM | $0.9241 \pm 0.0333$ | $0.9152 \pm 0.0351$ | $0.8769 \pm 0.0366$ | $0.8726 \pm 0.0256$ |
| Dataset 3 | AdaBoost | $0.8324 \pm 0.0030$ | $0.3681 \pm 0.0074$ | $0.7743 \pm 0.0105$ | $0.3825 \pm 0.0027$ |
| | XGBoost | $0.8386 \pm 0.0027$ | $0.3800 \pm 0.0053$ | $0.8016 \pm 0.0049$ | $0.3814 \pm 0.0020$ |
| | LightGBM | $0.8381 \pm 0.0026$ | $0.3801 \pm 0.0057$ | $0.7970 \pm 0.0053$ | $0.3825 \pm 0.0021$ |

Table 2: Five-fold cross-validation performance (mean ± standard deviation).

exceeded AdaBoost on ROC-AUC, PR-AUC, and Recall. The lower F1 scores of XGBoost and LightGBM indicate that their higher recall at the fixed 0.50 threshold came with a precision trade-off.

On Dataset 2, all three models achieved strong discrimination, with mean ROC-AUC values close to 0.924. XGBoost achieved the highest mean Recall and F1, whereas AdaBoost obtained the highest mean PR-AUC by a small margin. This shows that no single model dominated every metric on this relatively small dataset.

On Dataset 3, XGBoost and LightGBM again improved ROC-AUC and PR-AUC over AdaBoost. XGBoost achieved the highest mean Recall, while the three models had very similar F1 values. Across the three datasets, the aggregated comparison ranked XGBoost first with a mean dataset rank of 1.6667, followed by LightGBM at 2.0000 and AdaBoost at 2.3333. XGBoost also achieved the highest cross-dataset mean ROC-AUC (0.8834), PR-AUC (0.5717), and Recall (0.8205). AdaBoost achieved the highest cross-dataset mean F1 (0.5389), emphasizing that the preferred model depends partly on the evaluation objective and decision threshold.

| Model | Mean Rank | ROC-AUC | PR-AUC | Recall | F1 |
|---|---|---|---|---|---|
| XGBoost | **1.6667** | **0.8834** | **0.5717** | **0.8205** | 0.5308 |
| LightGBM | 2.0000 | 0.8832 | 0.5694 | 0.8170 | 0.5299 |
| AdaBoost | 2.3333 | 0.8760 | 0.5587 | 0.7662 | **0.5389** |

Table 3: Unweighted cross-dataset comparison based on the experiment outputs.

These results answer RQ1 by showing that XGBoost provides the strongest overall performance under the selected protocol, although LightGBM is extremely close on several ranking metrics and AdaBoost retains a small advantage in aggregate F1. Therefore, the comparison does not support a claim that one algorithm is uniformly superior on every metric and dataset.

## VI.2. Predictive Stability Across Cross-Validation Folds (RQ2)

The fold-level results also reveal clear differences in stability across the three datasets. Dataset 1 and Dataset 3 produced small standard deviations for all three models. For example, XGBoost obtained ROC-AUC standard deviations of 0.0033 on Dataset 1 and 0.0027 on Dataset 3. LightGBM showed similarly small variation, indicating that the two gradient-boosted models were not highly sensitive to the particular fold assignment on the two larger datasets.

Dataset 2 exhibited noticeably greater fold-to-fold variation. Its development partition contains

| Dataset | Model | ROC-AUC | PR-AUC | Recall | F1 |
|---|---|---|---|---|---|
| Dataset 1 | AdaBoost | 0.8695 | 0.3860 | 0.6705 | 0.3657 |
| | XGBoost | 0.8850 | 0.4169 | 0.7712 | 0.3255 |
| | LightGBM | 0.8850 | 0.4187 | 0.7694 | 0.3259 |
| Dataset 2 | AdaBoost | 0.9256 | 0.9394 | 0.8529 | 0.8832 |
| | XGBoost | 0.9224 | 0.9216 | 0.8333 | 0.8586 |
| | LightGBM | 0.9189 | 0.9166 | 0.8431 | 0.8643 |
| Dataset 3 | AdaBoost | 0.8326 | 0.3672 | 0.7674 | 0.3788 |
| | XGBoost | 0.8384 | 0.3756 | 0.7963 | 0.3773 |
| | LightGBM | 0.8385 | 0.3763 | 0.7925 | 0.3789 |

Table 4: Performance on the untouched hold-out test partitions after refitting on the full development data.

only 734 samples, compared with 353,653 for Dataset 1 and 183,824 for Dataset 3. XGBoost ROC-AUC ranged from 0.8676 to 0.9465 across the five folds, while AdaBoost ranged from 0.8547 to 0.9673 and LightGBM ranged from 0.8651 to 0.9497. The larger variability is therefore consistent with the smaller sample size: each validation fold contains fewer observations, so the composition of a single fold has a greater influence on the measured score.

The final hold-out results were generally close to the corresponding cross-validation means. For example, Dataset 1 XGBoost achieved a mean CV ROC-AUC of 0.8872 and a hold-out ROC-AUC of 0.8850, while Dataset 3 XGBoost achieved 0.8386 in CV and 0.8384 on the hold-out set. This agreement provides additional evidence that the reported cross-validation estimates are not driven by a single unusually favorable split.

Overall, RQ2 is answered by finding that predictive performance is highly stable on the two large datasets but less stable on the small Heart Failure Prediction dataset. The pattern is observed across all three algorithms, suggesting that dataset size and fold composition contribute more to the observed instability than the choice among the three boosting methods alone.

## VI.3. SHAP Feature-Importance Stability Across Cross-Validation Folds (RQ3)

Following the fold-wise explainability procedure described in Section V.4., RQ3 evaluates whether SHAP-based global feature-importance rankings remain consistent when the training and validation partitions change across cross-validation folds. For every dataset–model combination, all ten unique pairs of the five folds are compared using Kendall's $\tau$, Spearman's $\rho$, and top-10 Jaccard similarity.

Table 5 shows that the SHAP feature rankings are generally stable across folds for all nine dataset–model combinations. Mean Spearman correlations range from 0.9332 to 0.9700, indicating strong agreement in the complete feature ordering. Mean Kendall coefficients range from 0.8004 to 0.9599, providing additional evidence that most pairwise feature ordering relationships are preserved across folds. Top-10 Jaccard similarity ranges from 0.7455 to 1.0000, indicating that the most influential feature sets are also largely consistent.

Dataset 3 demonstrates particularly strong explanation stability. AdaBoost and XGBoost both obtain a mean top-10 Jaccard score of 1.0000, indicating that the same set of ten most influential features is retained across every fold pair. LightGBM achieves the highest overall mean Spearman correlation of 0.9700, showing that its complete feature ranking is also highly consistent. The

| Dataset | Model | Kendall $\tau$ | Spearman $\rho$ | Top-10 Jaccard |
|---|---|---|---|---|
| Dataset 1 | AdaBoost | 0.9599 | 0.9652 | 0.9273 |
| | XGBoost | 0.8004 | 0.9332 | 0.8545 |
| | LightGBM | 0.8198 | 0.9479 | 0.7455 |
| Dataset 2 | AdaBoost | 0.8562 | 0.9467 | 0.8061 |
| | XGBoost | 0.8222 | 0.9389 | 0.8727 |
| | LightGBM | 0.8170 | 0.9348 | 0.8364 |
| Dataset 3 | AdaBoost | 0.9130 | 0.9599 | **1.0000** |
| | XGBoost | 0.9004 | 0.9686 | **1.0000** |
| | LightGBM | 0.8988 | **0.9700** | 0.8727 |

Table 5: Mean SHAP feature-ranking stability across the ten pairwise comparisons of five cross-validation folds.

high Kendall coefficients for all three models further confirm that the relative ordering of feature pairs changes only modestly across partitions.

Dataset 1 also exhibits strong overall stability, although the different metrics reveal some variation near the top of the ranking. AdaBoost achieves the strongest agreement on this dataset, with mean Kendall $\tau = 0.9599$, Spearman $\rho = 0.9652$, and top-10 Jaccard similarity of 0.9273. LightGBM, however, obtains a high Spearman correlation of 0.9479 while its top-10 Jaccard similarity is lower at 0.7455. This result indicates that the complete ranking remains similar across folds even though several features near the top-10 boundary exchange positions or move into and out of the top feature set.

The difference between the Kendall and Spearman measurements is also informative. For example, Dataset 1 XGBoost achieves Spearman $\rho = 0.9332$ but Kendall $\tau = 0.8004$. Both values indicate positive and strong ranking agreement, but Kendall's lower value suggests that some pairwise feature orderings change across folds even though the overall ranking structure remains similar. Reporting both statistics therefore provides a more complete view of explanation robustness than relying on a single correlation measure.

Dataset 2 remains reasonably stable despite being substantially smaller than Datasets 1 and 3. All three models obtain mean Spearman correlations greater than 0.93, while mean Kendall coefficients range from 0.8170 to 0.8562. Top-10 Jaccard similarity ranges from 0.8061 to 0.8727. These values indicate that the majority of important features remain consistently ranked even though some movement occurs between folds.

The SHAP beeswarm plots complement the global importance rankings by showing both the magnitude and direction of feature contributions. Features appearing near the top of the beeswarm plots have large absolute effects on the model output, while the horizontal distribution of SHAP values indicates whether particular feature values tend to increase or decrease the predicted risk. The corresponding mean absolute SHAP bar plots provide a simpler view of global importance, and the violin plots show the distribution of SHAP effects for the leading features.

The feature-ranking heatmaps provide a direct visualization of explanation stability across folds. Features whose heatmap cells remain at similar rank values across all five folds can be considered highly stable, whereas large changes in rank indicate sensitivity to the particular training partition. This visualization is especially useful for distinguishing small changes near the top-10 cutoff that may reduce Jaccard similarity without substantially changing the complete ranking.

The fold-correlation heatmaps provide an additional model-level view of stability. High off-diagonal Spearman correlations indicate that different training folds produce similar global feature orderings. A fold with substantially lower correlations to the other folds would suggest that the corresponding training partition produces a noticeably different explanation. Together with the ranking heatmaps, these plots allow the numerical stability scores to be visually inspected rather than interpreted only from aggregated statistics.

Overall, RQ3 is answered by finding that SHAP-based global feature-importance rankings are robust to cross-validation partitioning for the evaluated boosting models. This conclusion is supported by three complementary measures: Kendall's $\tau$ shows strong pairwise ordering agreement, Spearman's $\rho$ shows high consistency in complete feature rankings, and top-10 Jaccard similarity shows that the most influential feature sets are generally preserved.

At the same time, the results demonstrate why multiple stability metrics are necessary. A model can maintain a high full-ranking correlation while still showing changes in the membership of its top-10 features. Consequently, explanation stability should not be judged from a single correlation coefficient alone. Combining rank correlations, top-feature overlap, feature-ranking heatmaps, and fold-correlation heatmaps provides a more complete assessment of the reproducibility of model explanations.

The observed SHAP stability should be interpreted as evidence of explanation robustness rather than causal validity. A feature that is consistently important across folds is repeatedly used by the model under the observed data distribution, but this does not imply that changing that feature would causally alter a patient's health outcome. In addition, because AdaBoost uses permutation SHAP while XGBoost and LightGBM use TreeSHAP, raw SHAP magnitudes are not used for direct cross-model comparisons. RQ3 therefore focuses on within-model fold-to-fold ranking stability, which remains comparable across the experimental protocol.

## VI.4. Discussion

Taken together, the three research questions provide complementary evidence about predictive quality and interpretability. RQ1 shows that XGBoost provides the strongest overall predictive ranking across the three datasets, with LightGBM very close on ROC-AUC and PR-AUC and AdaBoost retaining the highest cross-dataset mean F1. RQ2 shows that predictive scores are highly stable on the two large datasets but vary more on the smaller Heart Failure Prediction dataset. RQ3 extends the stability analysis from prediction metrics to model explanations and finds that the SHAP feature rankings remain highly consistent across folds for all three boosting methods.

The SHAP results should be interpreted as measures of explanation stability, not as evidence of causality. A feature can be consistently important to a model because it is predictive under the observed data distribution, but this does not establish that changing the feature would causally change a patient's risk. In addition, raw SHAP magnitudes are not compared directly between AdaBoost and the two gradient-boosted models because AdaBoost is explained with permutation SHAP whereas XGBoost and LightGBM use TreeSHAP. The study therefore bases RQ3 on within-model fold-to-fold ranking consistency, which is comparable across the experimental protocol.

Overall, the combined findings indicate that the strongest models are not only competitive in predictive performance but also produce feature-importance patterns that are reproducible across data partitions. This is useful for healthcare risk-prediction studies because an explanation that changes sharply with a different fold would be difficult to trust even if average predictive performance were high. The observed stability therefore provides an additional criterion, alongside ROC-AUC, PR-AUC, Recall, and F1, for assessing the robustness of the evaluated models.

# VII. Conclusion and Future Work

## VII.1. Conclusion

# VIII. References

[1] Y. Freund et al., "A decision-theoretic generalization of on-line learning and an application to boosting", *Journal of Computer and System Sciences*, vol. 55, no. 1, pp. 119–139, 1997. DOI: `10.1006/jcss.1997.1504`

[2] T. Chen et al., "XGBoost: A scalable tree boosting system", in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, ACM, 2016, pp. 785–794. DOI: `10.1145/2939672.2939785`

[3] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree", in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 3146–3154.

[4] S. M. Lundberg et al., "A unified approach to interpreting model predictions", in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[5] S. M. Lundberg et al., "From local explanations to global understanding with explainable ai for trees", *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020. DOI: `10.1038/s42256-019-0138-9`

[6] F. Pedregosa et al., "Scikit-learn: Machine learning in python", *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[7] T. Saito et al., "The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets", *PLOS ONE*, vol. 10, no. 3, e0118432, 2015. DOI: `10.1371/journal.pone.0118432`

# Appendix

1. **AIVIETNAM:** Further information about AIVIETNAM can be found here.
2. **GitHub Repository:**
3. **Dataset:**
4. **Demo Video / Presentation:**
    - Link demo video: Link demo
    - Link slide presentation: Link slide presentation